Challenge 1: Develop Your Problem Statement

Team Name: AgriClimate Innovators **Date: 10/10/2026**

A value-led problem statement clearly defines the core issue your team is addressing, why it matters, and who is most impacted. It anchors the AI development process in human-centric needs rather than pure technical capability.

1. The Core Problem

What specific problem are you trying to solve? Describe the current state and the negative impact it has.

Smallholder farmers in sub-Saharan Africa are facing increasingly unpredictable weather patterns due to climate change. Traditional ecological knowledge and seasonal calendars are no longer reliable predictors for planting and harvesting. This unpredictability leads to severe crop failures, reduced yields, and food insecurity, pushing vulnerable farming communities deeper into poverty.

2. Target Audience & Beneficiaries

Who experiences this problem? Be specific about demographics, contexts, and specific communities.

The primary beneficiaries are smallholder farmers in rural East and Southern Africa, specifically those managing plots under 2 hectares. Secondary beneficiaries include local agricultural extension workers and community cooperatives who support these farmers.

3. Proposed Value-Led Solution

How will your AI/SLM solution address this problem while upholding ethical values?

We are developing a Small Language Model (SLM) tailored to provide localized, climate-resilient agricultural advice. By processing regional weather forecasts, soil data, and historical crop performance, the SLM will generate actionable recommendations (e.g., optimal planting windows, pest mitigation). To ensure accessibility, the solution will be deployed via low-bandwidth SMS/USSD platforms, empowering farmers with critical insights without requiring smartphones or high-speed internet.